PS011 - Authorised Attack-Surface Inventory and Change Monitor

Continuous, scope-controlled discovery of what you expose, and alerts on what changes

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3. Problem Description

Organisations cannot protect what they cannot see. Internet-facing assets such as domains, subdomains, open ports, services and certificates appear, change and disappear constantly, and security teams usually find out only after something is exposed. [Add your own context: sector, scale, a real example.]

Key Issues

- **Unknown assets:** shadow IT and forgotten hosts are missing from the inventory
- **Unnoticed change:** new open ports, expired certificates or new services go undetected
- **Scope and legality:** scanning must stay strictly within authorised targets
- **Alert fatigue:** raw scan output is noisy and hard to prioritise

Target Users & Impact

Stakeholder	Pain Point	Impact
Security / SOC team	No trusted, current asset list	Slow detection of exposure
IT / compliance owners	Cannot prove what was scanned and when	Audit and policy gaps

4. Solution Proposed

Cybernova proposes a monitor that only scans targets registered in an authorised scope list. It discovers assets, stores them in a normalised inventory, compares each new snapshot with the previous one, and raises prioritised alerts on meaningful changes.

Solution Overview

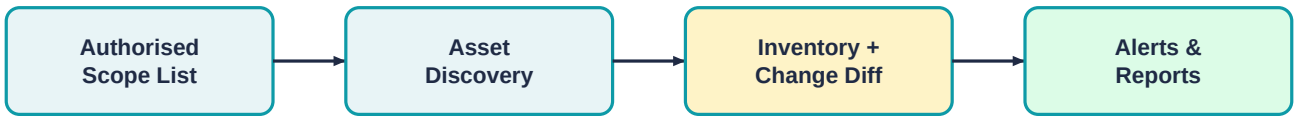

Figure 1: High-level solution flow

Key Features

- **Authorised scope registry:** every scan target is validated against approved domains and IP ranges, with an audit log
- **Asset discovery:** domains, subdomains, hosts, open ports, services and TLS certificates
- **Change detection:** snapshot diff showing new, removed and modified assets
- **Risk-ranked alerts and reports:** dashboard, notifications and exportable summaries

Technology Stack

Layer	Technologies
Frontend	[e.g., React dashboard]
Backend	[e.g., Python / FastAPI]
Database	[PostgreSQL / MongoDB / ...]
Discovery / Scanning	[e.g., DNS resolver, port scanner, TLS checker]
Deployment	[AWS / Vercel / Docker / ...]

5. Optimization Proposed by the Team

Optimizations & Enhancements

- **Optimization 1:** Incremental re-scans: only re-check assets that changed or are due
- **Optimization 2:** Hash-based snapshot comparison for fast diffs
- **Optimization 3:** Parallel scanning with rate limits to stay safe on targets
- **Enhancement:** alert de-duplication and severity scoring to cut noise

Before vs After Comparison

Parameter	Before Optimization	After Optimization	Improvement
Full scan time	[measured]	[measured]	[% faster]
Change-detection accuracy	[measured]	[measured]	[+ pts]
Alerts per scan (noise)	[measured]	[measured]	[% fewer]
Network requests / run	[measured]	[measured]	[% fewer]
Assets handled	[measured]	[measured]	[x]

Optimization Process Flowchart

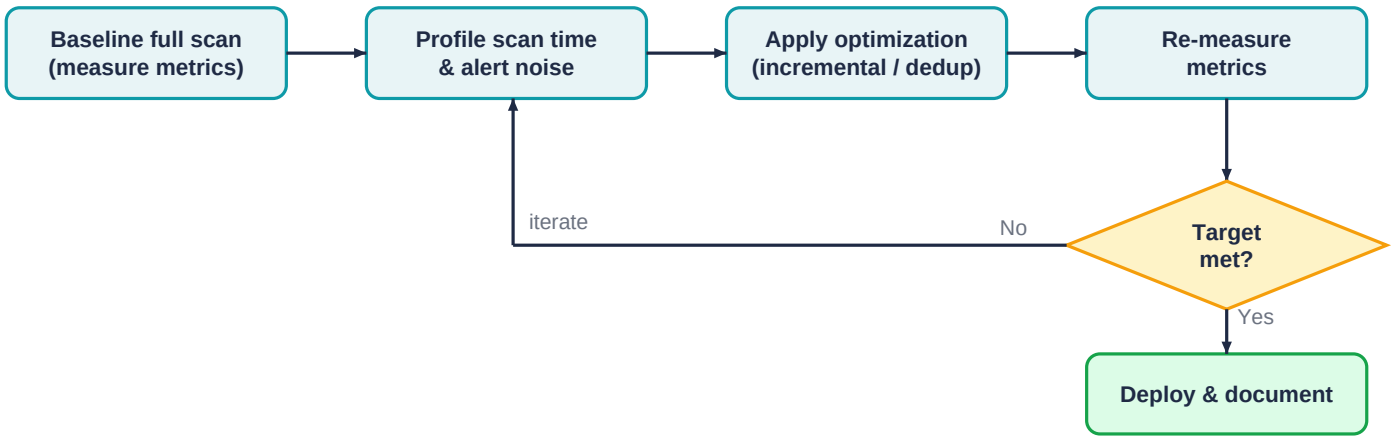

Figure 2: Iterative optimization process

6. Solution Architecture and Design

A Scope Guard validates every target before the Scheduler triggers discovery scanners. Results are normalised into the inventory database, the Change Detector diffs each snapshot against the last, and the Risk Scorer produces alerts for the dashboard and reports.

System Architecture Diagram

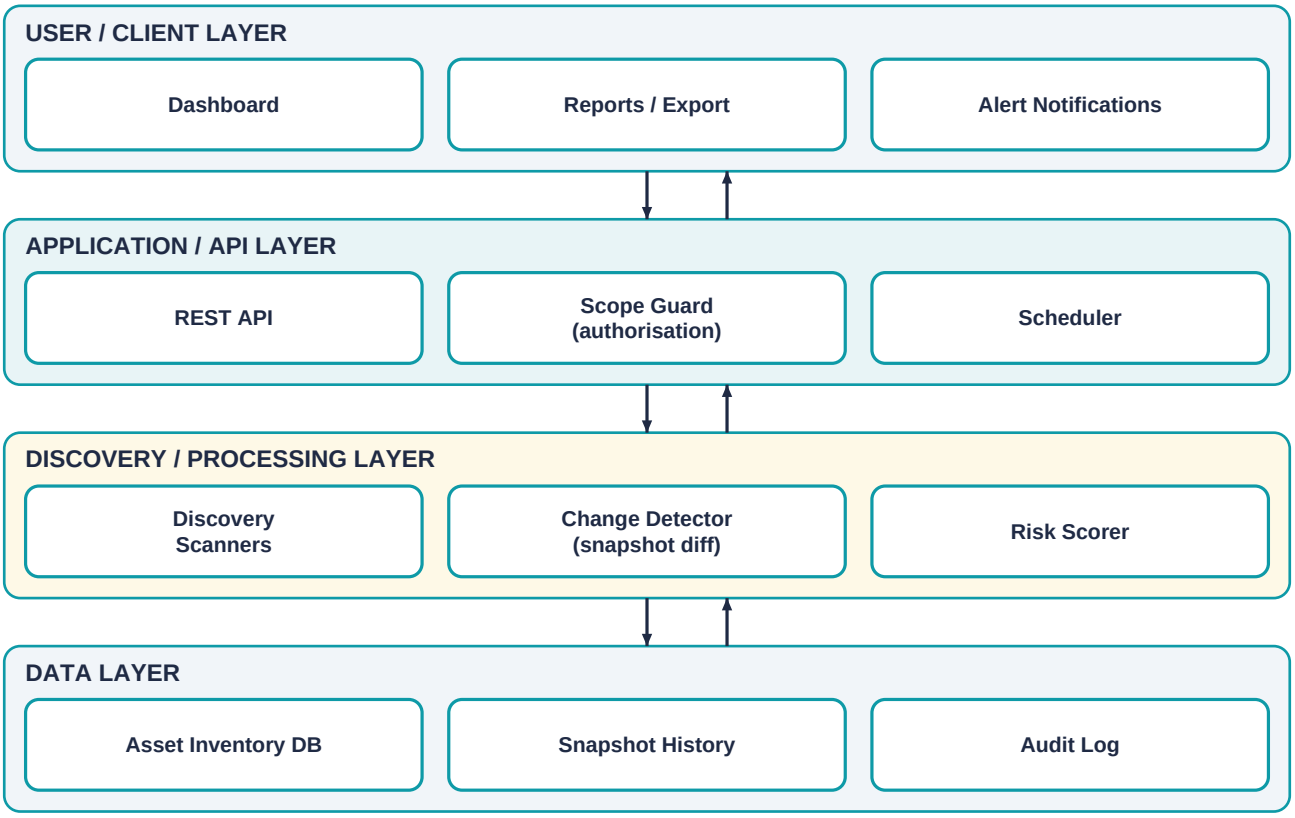

Figure 3: Layered solution architecture

Process Flow

Refer to **Figure 2** for the optimization flowchart and **Figure 1** for the solution flow. [Add sequence diagram / ER diagram / UI wireframes here if applicable.]

7. Timeline of Delivery

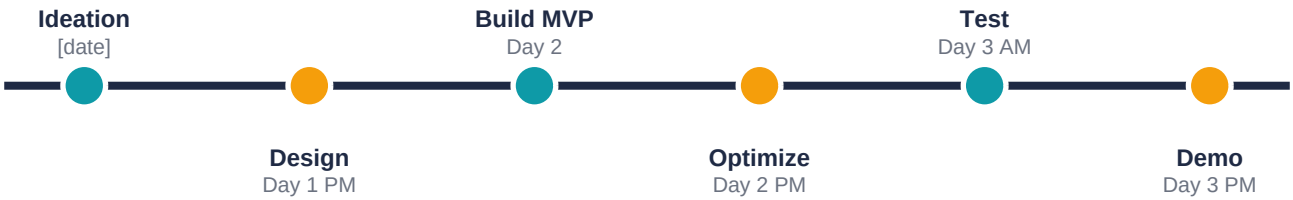

Figure 4: Delivery milestones

Milestone	Deliverable	Deadline	Owner
Ideation	Problem statement & scope	[Date/Time]	[Name]
Design	Architecture & wireframes	[Date/Time]	[Name]
Build MVP	Working core features	[Date/Time]	[Name]
Optimize	Performance & UX improvements	[Date/Time]	[Name]
Test	Bug fixes & validation	[Date/Time]	[Name]
Demo	Final pitch & submission	[Date/Time]	[Name]

8. References

Tools & Technologies

- [Tool / framework / library, with version]
- [Cloud / hosting service]
- [Design tool e.g., Figma, draw.io]

Datasets & APIs

- [Dataset name, source, link]
- [API documentation link]

Research & Reading

- [Author, Title, Publication, Year, URL]
- [Article / documentation link]

9. Conclusion

[Summarize the project: the problem, your solution, and its measurable results.]

Significance

- [Societal / business impact]
- [Scalability and future potential]

Next Steps / Call to Action

- [Pilot with real users / partners]
- [Add features: e.g., multilingual support, mobile app]
- [Seek mentorship, funding, or collaboration]